

(a) Case A: High-Risk Patient (True Positive)

Patient ID: 738291 | True Outcome: Adverse Event (Septic Shock)

Model Inputs (Patient Information)

STRUCTURED

- Age: 68 | Sex: M | BMI: 28.4
- Lactate: 4.2 mmol/L (High)
- Creatinine: 1.8 mg/dL (Elevated)
- SOFA Score: 7 (Critical)

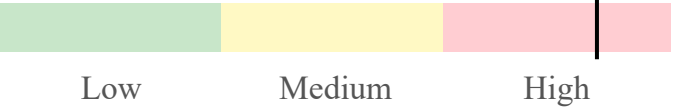
TEMPORAL

- HR: Sharp rise (90 → 125)
- MAP: Progressive decline (75 → 58)
- SpO2: Desaturation (96% → 91%)
- RR: Tachypnea (>24 sustained)

TEXTUAL

- Lethargy; pain-only response
- Mottled skin; warm extremities
- Rapid mental status decline
- Suspect sepsis; fluid initiated

Risk Assessment & Fusion
Risk Score: 0.89



- Structured
High Impact
- Temporal
Critical
- Textual
Supportive

Model Interpretation

The model assigns a High Risk (0.89) primarily driven by the 'Temporal' modality, detecting a classic shock trajectory. This is reinforced by 'Structured' markers of hypoperfusion. The 'Textual' encoder identifies keywords like 'lethargic' and 'mottled skin', confirming septic shock signals.

(b) Case B: Low-Risk Patient (True Negative)

Patient ID: 291038 | True Outcome: No Adverse Event (Discharge)

Model Inputs (Patient Information)

STRUCTURED

- Age: 54 | Sex: F | BMI: 24.1
- Lactate: 1.1 mmol/L (Normal)
- Creatinine: 0.9 mg/dL (Stable)
- SOFA Score: 1 (Minimal)

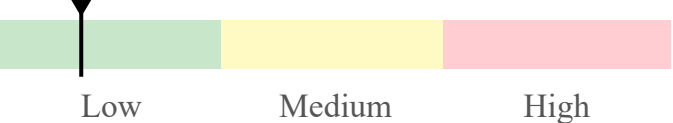
TEMPORAL

- HR: Stable (70-80 bpm)
- MAP: Maintained (>75 mmHg)
- SpO2: 98% (Room air)
- Temp: Afebrile

TEXTUAL

- Pain well-controlled
- Alert & oriented x3
- No acute distress
- Hemodynamically stable

Risk Assessment & Fusion
Risk Score: 0.12



- Structured
Low Risk
- Temporal
Stable
- Textual
Reassuring

Model Interpretation

The model predicts Low Risk (0.12) due to the absence of alarm signals. 'Temporal' analysis shows stable vital signs. 'Structured' inputs indicate normal organ function. Crucially, the 'Textual' analysis picks up negative sentiment for risk (e.g., 'well-controlled', 'stable'), ruling out deterioration.